(i) *Less than Cumulative Frequency Distribution:* In this type of cumulative frequency distribution, the cumulative frequency for each class shows the number of elements in the data whose magnitudes are less than the upper limit of the respective class.

(ii) *More than Cumulative Frequency Distribution:* In this type of cumulative frequency distribution, the cumulative frequency for each class shows the number of elements in the data whose magnitudes are larger than the lower limit of the class.

**Example 3.15**

Construct less than and more than cumulative frequency distribution tables for the following frequency distribution of orders received by a business firm over a number of weeks during a year.

| **Number of order received** | 0 - 4 | 5 - 9 | 10 - 14 | 15 - 19 | 20 - 24 |
|---|---|---|---|---|---|
| **Number of weeks** | 2 | 8 | 11 | 14 | 6 |
| **Number of order received** | 25 - 29 | 30 - 34 | 35 - 39 | 40 - 44 | 45 - 49 |
| **Number of weeks** | 4 | 3 | 2 | 1 | 1 |

***Solution:***

For the data related to the number of orders received per week by a business firm during a period of one year, the less than and more than cumulative frequencies are computed and are given in Table 3.15

Table 3.15

Cumulative Frequency Distribution for the number of orders received by a Business firm

| **Given data** | | **Less than ogive** | | **More than ogive** | |
|---|---|---|---|---|---|
| **Number of Orders Received** | **Number of Weeks** | **Upper limit** | **Less than Cumulative Frequencies** | **Lower limit** | **More Than Cumulative Frequencies** |
| 0 – 4 | 2 | 4 | 2 | 0 | 52 |
| 5 – 9 | 8 | 9 | 10 | 5 | 50 |
| 10 – 14 | 11 | 14 | 21 | 10 | 42 |
| 15 – 19 | 14 | 19 | 35 | 15 | 31 |
| 20 – 24 | 6 | 24 | 41 | 20 | 17 |
| 25 – 29 | 4 | 29 | 45 | 25 | 11 |
| 30 – 34 | 3 | 34 | 48 | 30 | 7 |



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